

$$G(\lambda) = \frac{32,4 \cdot \lambda + 3240}{(\lambda - 2,16) \cdot (\lambda + 2,31)}$$

$$G(z) = (1 - z^{-1}) \cdot \mathcal{Z}\left\{\frac{G(\lambda)}{\lambda}\right\}$$

$$\frac{G(\lambda)}{\lambda} = \frac{A}{\lambda} + \frac{B}{\lambda - 2,16} + \frac{C}{\lambda + 2,31}$$

$$A = ? \mid \lambda = 0$$

$$A = \frac{32,4 \cdot 0 + 3240}{(0 - 2,16) \cdot (0 + 2,31)} = \frac{3240}{(-4,9896)} = -649,3506$$

$$B = ? \mid \lambda = 2,16$$

$$B = \frac{32,4 \cdot \lambda + 3240}{\lambda \cdot (\lambda + 2,31)} = \frac{32,4 \cdot 2,16 + 3240}{2,16 \cdot (2,16 + 2,31)} = \frac{3309,984}{9,6552} = 342,8188$$

$$C = ? \mid \lambda = -2,31$$

$$C = \frac{32,4 \cdot \lambda + 3240}{\lambda \cdot (\lambda - 2,16)} = \frac{32,4 \cdot (-2,31) + 3240}{(-2,31) \cdot [(-2,31) - 2,16]} = \frac{3165,156}{10,3257} = 306,5318$$

$$\frac{G(\lambda)}{\lambda} = \frac{(-649,3506)}{\lambda} + \frac{342,8188}{(\lambda - 2,16)} + \frac{306,5318}{(\lambda + 2,31)}$$

$$\mathcal{Z}\left\{\frac{G(\lambda)}{\lambda}\right\} = \frac{(-649,3506)}{1 - z^{-1}} + \frac{342,8188}{1 - e^{2,16 \cdot 0,01} \cdot z^{-1}} + \frac{306,5318}{1 - e^{-2,31 \cdot 0,01} \cdot z^{-1}}$$

$$\mathcal{Z}\left\{\frac{G(n)}{n}\right\} = \frac{(-649,3506)}{1-z^{-1}} + \frac{342,8188}{1-1,0218 \cdot z^{-1}} + \frac{306,5318}{1-0,9772 \cdot z^{-1}}$$

$$(1-1,0218 \cdot z^{-1})(1-0,9772 \cdot z^{-1}) = 1 - 1,999 \cdot z^{-1} + 0,9985 \cdot z^{-2}$$

$$(1-z^{-1})(1-0,9772 \cdot z^{-1}) = 1 - 1,9772 \cdot z^{-1} + 0,9772 \cdot z^{-2}$$

$$(1-z^{-1})(1-1,0218 \cdot z^{-1}) = 1 - 2,0218 \cdot z^{-1} + 1,0218 \cdot z^{-2}$$

$$(-649,3506) \cdot (1 - 1,999 \cdot z^{-1} + 0,9985 \cdot z^{-2}) = (-649,3506) + 1298,0518 \cdot z^{-1} - 648,3766 \cdot z^{-2}$$

$$342,8188 \cdot (1 - 1,9772 \cdot z^{-1} + 0,9772 \cdot z^{-2}) = 342,8188 - 677,8213 \cdot z^{-1} + 335,0025 \cdot z^{-2}$$

$$306,5318 \cdot (1 - 2,0218 \cdot z^{-1} + 1,0218 \cdot z^{-2}) = 306,5318 - 619,7460 \cdot z^{-1} + 313,2142 \cdot z^{-2}$$

$$G(z) = \frac{\cancel{(1-z^{-1})} \cdot (0 + 0,4845 \cdot z^{-1} - 0,1599 \cdot z^{-2})}{\cancel{(1-z^{-1})} \cdot (1-1,0218 \cdot z^{-1})(1-0,9772 \cdot z^{-1})}$$

$$G(z) = \frac{0,4845 \cdot z^{-1} - 0,1599 \cdot z^{-2}}{1 - 1,999 \cdot z^{-1} + 0,9985 \cdot z^{-2}} \cdot \frac{z^2}{z^2} = \frac{\overbrace{0,4845 \cdot z - 0,1599}^{\rightarrow (0,4845 - 0,1599) \cdot z}}{z^2 - 1,999 \cdot z + 0,9985}$$

$$G(z) = \frac{0,3246 \cdot z}{z^2 - 1,999 \cdot z + 0,9985}$$